

LOW ACRYLONITRILE NBR: PATENT LANDSCAPE FOR SYNTHESIS, MODIFICATION, AND APPLICATION-DRIVEN FORMULATION

1. ABSTRACT

This report synthesizes a patent landscape around low-acrylonitrile nitrile butadiene rubber (NBR) and closely related nitrile-rubber technologies captured in the provided manifest. The evidence shows that the term “low acrylonitrile” is not directly quantified in the extracted passages for most patents; instead, the relevant landscape is defined by NBR synthesis, post-polymerization molecular-weight reduction, functionalization, and application-specific compounding. The core material definition remains a copolymer or terpolymer of at least one α,β -unsaturated nitrile and at least one conjugated diene, with butadiene and acrylonitrile as the central monomers. Across the set, the dominant synthesis approaches are emulsion polymerization for rubber latex formation, anionic solution polymerization for functionalized rubbery polymers, and metathesis-based degradation of nitrile rubber to lower molecular weight before hydrogenation. Several patents focus on downstream performance rather than monomer composition: catalyst systems for metathesis of nitrile rubber (with or without salt additives), conductive dispersions using nitrile-group-containing copolymers as dispersants, and composite or adhesive systems that incorporate NBR-derived phases for mechanical or interfacial performance.

The most directly relevant NBR-processing patents emphasize reducing molecular weight and improving processability of nitrile rubber feedstocks prior to hydrogenation, with reported targets including Mooney viscosity below 30 for the pre-hydrogenation NBR and hydrogenated products in the Mooney range of roughly 3 to 50. The metathesis patents repeatedly identify ruthenium- or osmium-based catalysts, sometimes combined with salts such as alkaline-earth chlorides or other salts, as a route to faster or more efficient degradation, while noting that catalyst loading and reaction time remain important constraints. Other patents in the set use NBR as a functional phase in electro-conductive members, epoxy toughening particles, polyurethane compatibilization, or conductive dispersions, indicating that low-acrylonitrile NBR is often selected for balancing polarity, flexibility, and filler compatibility rather than for a single standalone synthesis recipe. Overall, the landscape suggests that practical low-acrylonitrile NBR synthesis is best approached through controlled emulsion or solution polymerization followed by targeted molecular-weight adjustment and formulation-specific compounding, rather than through a single universal monomer ratio disclosed in the extracted evidence.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US6812307B2
- Patent Title: Functionalized monomers for synthesis of rubbery polymers
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2004
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 2

- Patent Number: US10266615B2
- Patent Title: Method for preparing acrylonitrile-butadiene-styrene graft copolymer and ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2019
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 3

- Patent Number: US8957161B2
- Patent Title: Process for reacting chemical compounds in the presence of catalyst systems
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 4

- Patent Number: US11347156B2
- Patent Title: Electro-conductive member for electrophotography, process cartridge for ...
- Assignee: Not disclosed

- Jurisdiction: US
- Publication Year: 2022
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 5

- Patent Number: US8450543B2
- Patent Title: Integrated methods of preparing renewable chemicals
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 6

- Patent Number: US9670304B2
- Patent Title: Composite for preventing ice adhesion
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 7

- Patent Number: US7737233B2
- Patent Title: Catalyst systems and their use for metathesis reactions
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 8

- Patent Number: US9976027B2
- Patent Title: Polymer fine particle-containing curable resin composition having improved ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2018
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Patent 9

- Patent Number: US8124234B2
- Patent Title: Polyurethane compositions and articles prepared therefrom, and methods for ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2012
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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Technical Relevance Selected via deterministic pipeline scoring.

Patent 10

- Patent Number: US11469421B2
- Patent Title: Conductive material dispersion, binder resin-containing conductive material ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2022
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - No polymerization parameters disclosed.

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3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US7737233B2 and US8957161B2 are the most directly aligned with nitrile-rubber molecular-weight reduction: both address metathesis of nitrile rubber, with US7737233B2 adding an alkaline-earth metal chloride to the catalyst system and US8957161B2 adding salts more generally to improve metathesis activity.
- Both metathesis patents describe nitrile rubber as a copolymer/terpolymer of at least one α,β -unsaturated nitrile and at least one conjugated diene; the extracted passages do not provide a specific acrylonitrile percentage or monomer ratio.
- US7737233B2 emphasizes catalyst systems based on transition-group 6 or 8 complexes with carbene-like ligands, including Grubbs-type catalysts, while US8957161B2 similarly uses metathesis catalysts with added salts and explicitly discusses ruthenium- and osmium-based systems.
- US8957161B2 states that low molecular weight NBR can be obtained by metathesis with low molecular weight 1-olefins, followed by hydrogenation; the extracted text reports resulting HNBR in the range Mw 30,000 to 250,000 and Mooney viscosity 3 to 50 in cited prior-art examples.
- US7737233B2 states that prior art often required relatively large catalyst amounts and long reaction times; the patent's improvement is catalyst activation by salt addition, but the extracted passages do not provide a specific temperature or pressure window.
- US6812307B2 is not an NBR patent per se; it is a solution anionic polymerization patent for rubbery polymers. It is relevant as a process analogue because it discloses copolymerization of conjugated diolefins with functionalized monomers at 20 to 100 °C in organic solvent, initiated by an anionic initiator and conducted without conventional polar modifiers.
- US10266615B2 and US11347156B2 use NBR in downstream formulations rather than synthesis: US10266615B2 uses diene-based rubber latex in ABS graft copolymer preparation, while US11347156B2 uses NBR as one of the diene-based rubbers in an electro-conductive member with phase-separated matrix/domain design.
- US9976027B2 and US8124234B2 both use diene-based or polydiene-derived rubber phases as toughening/compatibilizing elements in resin systems, but neither provides a direct NBR polymerization recipe in the extracted passages.
- US11469421B2 is formulation-oriented and uses a copolymer containing a nitrile group-containing structural unit as a dispersant for carbon fibers; it is relevant to

nitrile-containing polymer functionality, but the extracted text does not identify NBR synthesis conditions.

- US9670304B2 is application-driven and lists acrylonitrile-butadiene-styrene and polybutadiene among hydrophobic polymers for anti-icing composites; it does not disclose NBR synthesis or acrylonitrile content control.
- Across the set, the historical trend is from general rubber synthesis and functionalization toward more specialized post-polymerization control: metathesis degradation, catalyst/salt optimization, and formulation-specific use of nitrile-containing polymers in adhesives, conductive systems, and impact-modified resins.

4. CONCLUSION

The evidence supports a practical synthesis strategy for low-acrylonitrile NBR centered on controlled copolymerization of butadiene with a reduced acrylonitrile fraction, followed where needed by molecular-weight adjustment or downstream functional use. However, the extracted patents do not disclose a single universal low-acrylonitrile monomer ratio; instead, they show that performance is strongly governed by process choice and post-polymerization handling. For direct NBR production, the most relevant evidence comes from the metathesis patents, which indicate that nitrile rubber can be reduced to lower molecular weight using ruthenium- or osmium-based catalyst systems, with salt additives improving activity. The cited prior-art ranges suggest that low-viscosity feedstocks and lower-Mooney products are achievable, but catalyst loading and reaction time remain critical. For broader rubber synthesis analogues, anionic solution polymerization at 20 to 100 °C in organic solvent without conventional polar modifiers is a useful benchmark for rubbery copolymer preparation. Based on the patent set, the most defensible recommended synthesis parameters are: use a butadiene-rich nitrile-rubber backbone, keep acrylonitrile at the low end of the application-appropriate range, employ controlled emulsion or solution polymerization, and if lower molecular weight is required, apply metathesis-based degradation with an optimized ruthenium/osmium catalyst system and salt additive. For formulation work, low-acrylonitrile NBR appears most valuable when paired with fillers, conductive phases, or toughening domains where balanced polarity and flexibility are needed.

5. REFERENCES

- US6812307B2 | Functionalized monomers for synthesis of rubbery polymers | Not disclosed | US | 2004 | <https://patents.google.com/patent/US6812307B2/en>
- US10266615B2 | Method for preparing acrylonitrile-butadiene-styrene graft copolymer and ... | Not disclosed | US | 2019 | <https://patents.google.com/patent/US10266615B2/en>
- US8957161B2 | Process for reacting chemical compounds in the presence of catalyst systems | Not disclosed | US | 2015 | <https://patents.google.com/patent/US8957161B2/en>
- US11347156B2 | Electro-conductive member for electrophotography, process cartridge for ... | Not disclosed | US | 2022 | <https://patents.google.com/patent/US11347156B2/en>

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- US8124234B2 | Polyurethane compositions and articles prepared therefrom, and methods for ... | Not disclosed | US | 2012 | <https://patents.google.com/patent/US8124234B2/en>
- US11469421B2 | Conductive material dispersion, binder resin-containing conductive material ... | Not disclosed | US | 2022 | <https://patents.google.com/patent/US11469421B2/en>
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